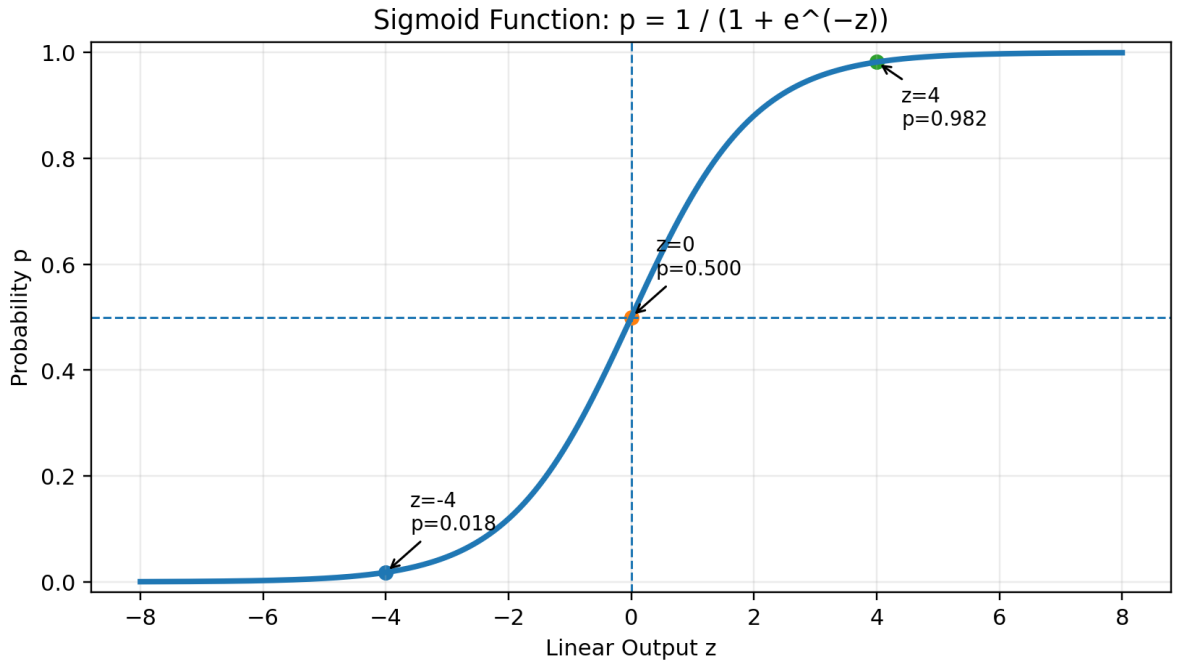
Logistic Regression Notes

Student-friendly explanation based on the 8-slide PPT, with added images and formulas



What you will learn: intuition of logistic regression, why a straight line is not suitable for binary output, the sigmoid function, log-odds, predicted probabilities, and threshold-based classification.

This PDF is written as class notes, so every idea is explained in simple language and arranged in the same order as the PPT slides.

1. Logistic Regression Intuition

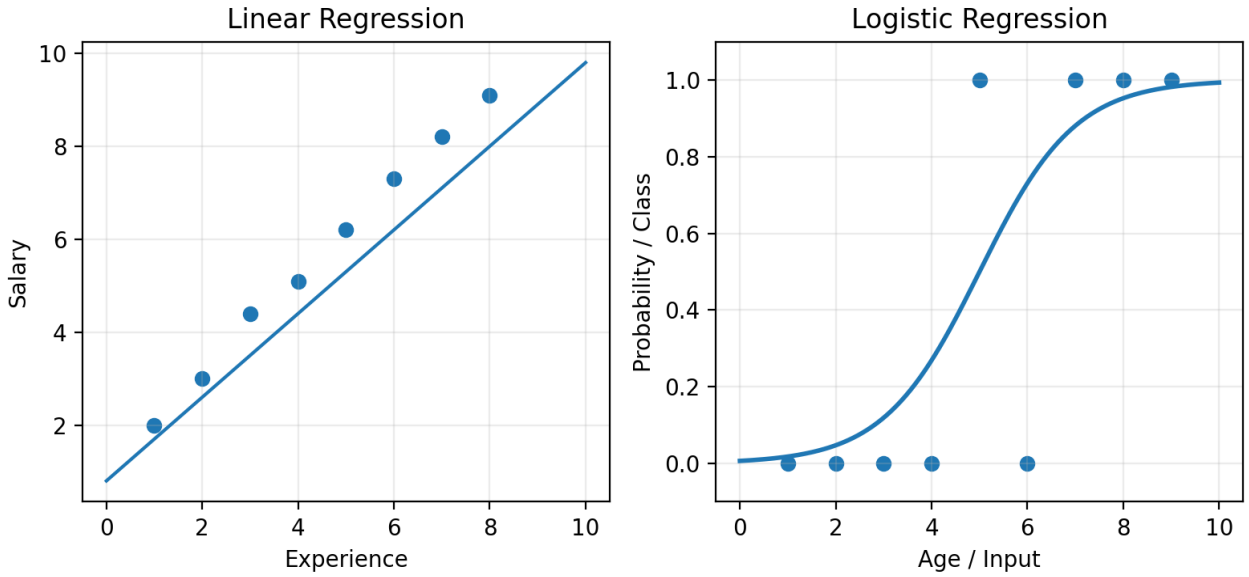
Logistic Regression is a supervised machine learning algorithm used when the output has only two possible classes, such as Yes/No, Pass/Fail, Positive/Negative, or Spam/Not Spam. Instead of directly giving a category, the model first calculates a **probability**. That probability is then converted into a final class label.

In simple words, logistic regression answers a question like: “What is the chance that this input belongs to class 1?”

Important note: despite the word “regression” in its name, logistic regression is mainly used for **classification**.

2. Linear Regression vs Logistic Regression

The PPT compares two kinds of prediction problems. In linear regression, the output is a continuous numerical value, such as salary, marks, or temperature. In logistic regression, the output is binary, which means it can take only 0 or 1.



For linear regression, the equation is:

$$y = b_0 + b_1 x$$

Here, b_0 is the intercept, b_1 is the slope or weight, and x is the input feature. A straight line works well when the output is numerical, but it becomes problematic when the output should behave like a probability.

3. Why a Straight Line Is Not Suitable

A straight line can produce any value from minus infinity to plus infinity. For example, a line may predict –0.2 for one input and 1.3 for another input. These values cannot be treated as valid probabilities because probability must always lie between 0 and 1.

$$0 \leq p \leq 1$$

So, for binary classification, we need a mathematical function that keeps the output within this valid range. That is why logistic regression uses the sigmoid function.

4. Restricting the Output to 0–1

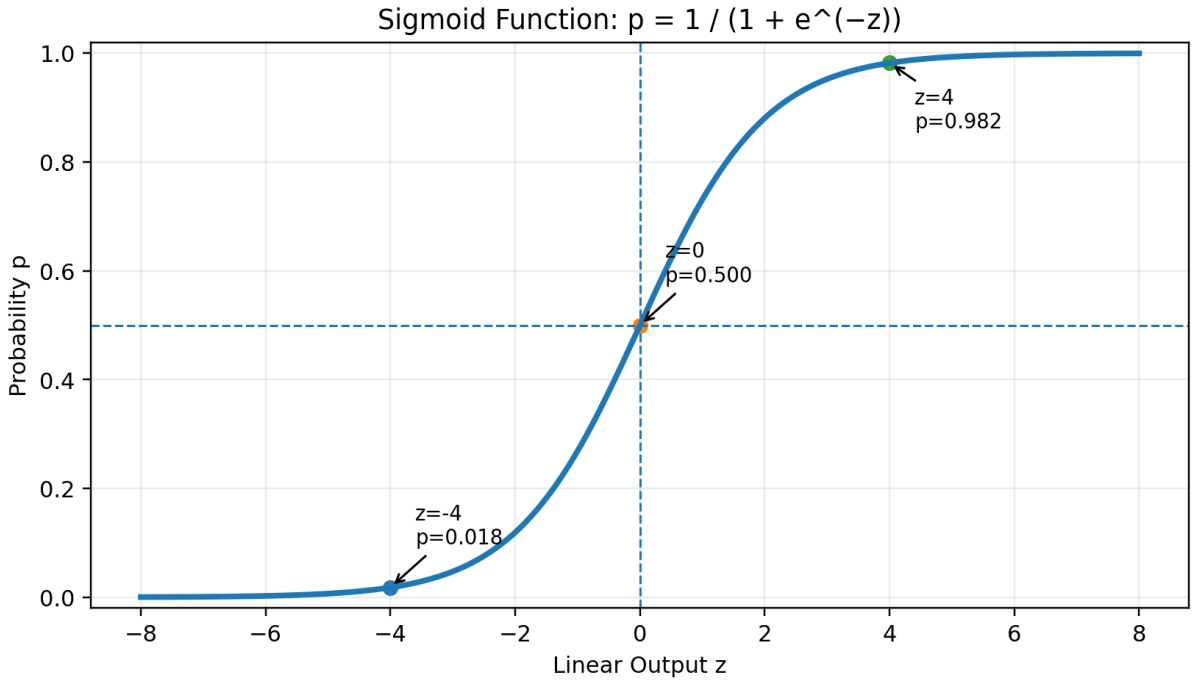
One idea is to force the output to stay between 0 and 1 by cutting off values below 0 and above 1. But that creates a graph with sharp corners, which is not smooth and not mathematically convenient for learning. Instead, logistic regression uses a smooth S-shaped curve.

Desired behavior of the curve:

- It should stay close to 0 for very small input values.
- It should rise gradually in the middle region.
- It should approach 1 for large input values.
- It should never cross below 0 or above 1.

5. Sigmoid Curve and Log-Odds

The sigmoid curve is an S-shaped curve that converts any real-valued input into a probability. This is the main idea behind logistic regression.



The complete sigmoid function is:

$$\sigma(z) = 1 / (1 + e^{-z})$$

Often, the variable z represents the linear output of the model. For one input feature, $z = b_0 + b_1x$. For multiple features, $z = b_0 + b_1x_1 + b_2x_2 + \ldots + b_nx_n$.

The logistic regression model can therefore be written as:

$$P(Y = 1 \mid X) = 1 / (1 + e^{-(b_0 + b_1x_1 + b_2x_2 + \ldots + b_nx_n)})$$

The PPT also shows the equivalent **log-odds** form:

$$\ln(p / (1 - p)) = b_0 + b_1x$$

Here, p is the probability of class 1, 1 – p is the probability of class 0, p/(1 – p) is called the odds, and ln is the natural logarithm.

Important point: logistic regression makes the **log-odds** linear, while the probability itself follows a sigmoid curve.

6. From Linear Output to Probability

The model works in two simple steps:

1. First compute the linear score z = b0 + b1x (or a multi-feature version).
2. Then pass that score through the sigmoid function to get p.

Small numerical example:

$$\text{If } z = 2, \text{ then } p = 1 / (1 + e^{-2}) = 1 / (1 + 0.1353) = 0.8808$$

So the model predicts an 88.08% chance of class 1. Similarly, if z = 0, then p = 0.5, which lies exactly at the middle of the sigmoid curve.

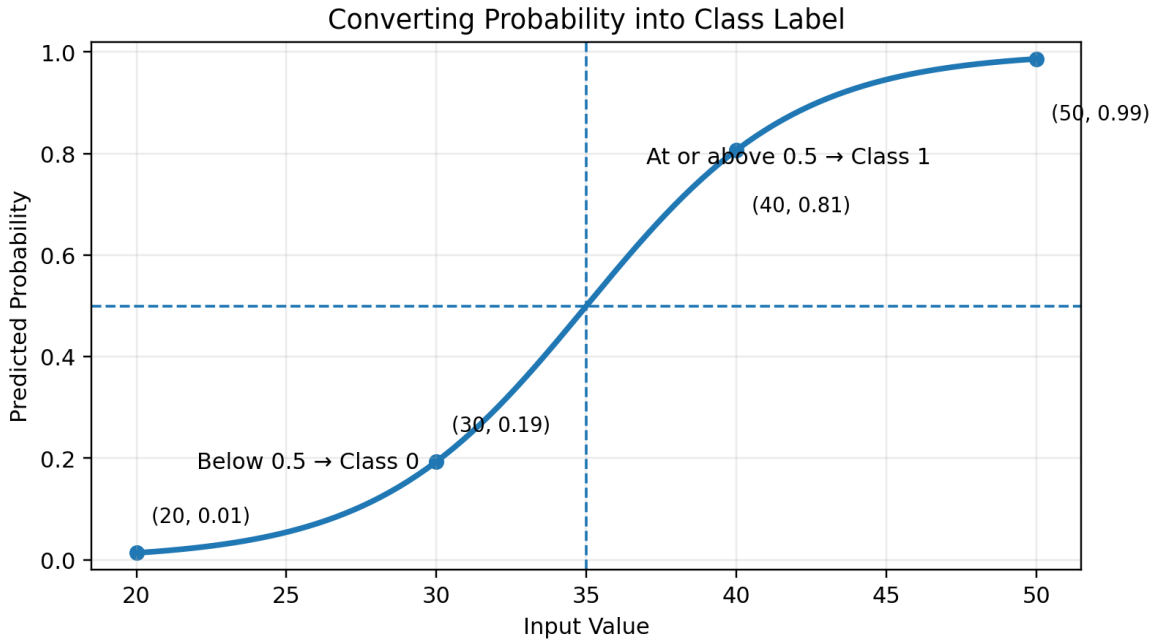
7. Understanding Predicted Probabilities

A predicted probability tells us how strongly the model believes that the input belongs to class 1. For example, p = 0.85 means there is an 85% predicted chance of class 1. This is still a probability—not yet the final label.

Input shown	Meaning on the curve
20	Very low probability
30	Probability has begun increasing
40	High probability
50	Probability is close to 1

8. Converting Probability into Class 0 or 1

To make a final prediction, the model compares the probability with a threshold. In most basic examples, the threshold is 0.5.



Classification rule:

■ = 0, if p < 0.5 and ■ = 1, if p ≥ 0.5

Predicted probability	Predicted class
0.07	0 — No
0.23	0 — No
0.50	1 — Yes
0.85	1 — Yes
0.994	1 — Yes

Important note: probability and predicted class are not the same thing. The threshold converts a probability into a final category.

9. Complete Flow of Logistic Regression

The complete working process can be summarized as follows:



Flow summary: linear equation → sigmoid function → probability → threshold decision → final class.

10. Final Takeaway

Logistic Regression begins with a linear equation, converts that linear output into a probability using the sigmoid function, and finally uses a threshold to predict class 0 or class 1. It is simple, interpretable, and widely used for binary classification problems.

Quick recap:

- Linear equation: z = b0 + b1x
- Sigmoid function: p = 1 / (1 + e^(–z))
- Probability range: 0 ≤ p ≤ 1
- Decision rule: if p ≥ 0.5, predict class 1; otherwise predict class 0

These notes are an explanatory version of the PPT, written for easy student understanding. The examples and diagrams are supportive learning material added to make the topic self-explanatory.